

CBSE Practice Worksheet

Class 8 | Science | Difficulty: Hard | Set B

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
2. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
3. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
4. What is the hybridization of carbon in methane?
(A) sp
(B) sp²
(C) sp³
(D) sp³d
5. What is the oxidation state of Mn in KMnO₄?
(A) +2
(B) +4
(C) +6
(D) +7
6. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
7. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis
(D) Respiration
8. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1

(D) 0 only

9. Which law relates induced EMF to rate of change of magnetic flux?

- (A) Lenz's law
- (B) Faraday's law
- (C) Ampere's law
- (D) Ohm's law

10. Which quantum number determines the shape of an orbital?

- (A) Principal
- (B) Azimuthal
- (C) Magnetic
- (D) Spin

Section B: Fill in the Blanks (1 mark each)

- 11. For a reversible process, entropy change of the universe is ____.
- 12. The uncertainty principle was given by ____.
- 13. The enzyme that converts prothrombin to thrombin is ____.
- 14. In SN2 reactions, the rate depends on both ____ and nucleophile.
- 15. The process of splitting water during photosynthesis is called ____.
- 16. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 17. The band gap in semiconductors is typically ____ eV.
- 18. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.

Section C: Short Answer Questions (2 marks each)

- 19. Derive the equation for kinetic energy of gases from kinetic theory.
- 20. Describe electromagnetic induction and derive Faraday's law.
- 21. Explain the mechanism of DNA replication.
- 22. Explain the structure and working of a nuclear reactor.
- 23. Describe the light-dependent and light-independent reactions of photosynthesis.
- 24. Explain Le Chatelier's principle with examples.
- 25. Describe the mechanism of nerve impulse transmission.

--- End of Worksheet ---

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